

Confidence Intervals for Population Proportion

Statistics Worksheet · Grade 11–12 / Introductory College Statistics

Name: _____

Date: _____

Learning Objectives

- Identify the population and parameter of interest in a real-world context
- Verify the three conditions (randomness, independence, normality) required to construct a confidence interval for a population proportion
- Construct and interpret a confidence interval for a population proportion and determine the required sample size

Problems

1. A researcher surveys 150 randomly selected high school students and finds that 45 of them own a pet. Identify the population and the parameter of interest.

2. In a study, 200 college students were randomly selected and 80 said they exercise at least 3 times per week. Calculate the sample proportion $\hat{p}$ and the complement $\hat{q}$.

$$\hat{p} = \frac{x}{n}, \quad \hat{q} = 1 - \hat{p}$$

3. A study used a simple random sample of 172 undergraduate students at a large university. Check the three conditions needed to use a confidence interval for a population proportion, given that 19 students said they would report cheating. Show all work for the normality condition.

$$n\hat{p} \geq 10 \quad \text{and} \quad n\hat{q} \geq 10$$

4. Find the critical value z^* for a 95% confidence interval for a population proportion.

$$\alpha = 1 - 0.95, \quad \frac{\alpha}{2} = ?$$

5. A poll found that 136 out of 400 randomly selected adults prefer online shopping. Construct a 90% confidence interval for the true population proportion who prefer online shopping.

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$



6. Interpret the following confidence interval in context: A 95% confidence interval for the proportion of adults who support a new policy is (0.52, 0.68).

7. A researcher wants to estimate the proportion of voters who favor a new law. She wants a margin of error of no more than 0.03 with 95% confidence, and she has no prior estimate of the proportion. Find the minimum required sample size.

$$n = \left(\frac{z^*}{E}\right)^2 \cdot \hat{p}\hat{q}$$

8. Using the data from the video, a study found that $\hat{p} = 0.75$ with a 90% confidence level and a desired margin of error of 0.04. Derive the sample size formula from the margin-of-error formula and then compute the required sample size.

9. A sample of 250 randomly selected nurses found that 175 reported experiencing burnout. Construct a 99% confidence interval for the true proportion of nurses experiencing burnout. Then state whether the conditions are met and interpret the interval.

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

10. Two studies each estimate the proportion of college students who use social media daily. Study A uses $n = 100$ and gets $\hat{p} = 0.78$. Study B uses $n = 900$ and gets $\hat{p} = 0.78$. Both use 95% confidence. Without calculating, explain which study produces the narrower interval and why. Then calculate both margins of error to confirm your reasoning.

$$E = z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$



Confidence Intervals for Population Proportion — Answer Key

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Answer Key

1. Answer: Population: all high school students. Parameter of interest: the proportion of high school students who own a pet.

- The population is the entire group being studied: all high school students.
- The parameter of interest is what we are estimating: the true proportion (p) of high school students who own a pet.

2. Answer: $p\text{-hat} = 0.40$, $q\text{-hat} = 0.60$

- $p\text{-hat} = 80 / 200 = 0.40$
- $q\text{-hat} = 1 - 0.40 = 0.60$

3. Answer: All three conditions are satisfied: SRS confirmed; $N > 10(172) = 1720$ assumed; $np\text{-hat} = 19 \geq 10$ and $nq\text{-hat} = 153 \geq 10$.

- Randomness: The problem states an SRS of 172 students — condition satisfied.
- Independence: The university is large, so we can assume the total number of undergraduates $N > 10 \times 172 = 1720$ — condition satisfied.
- Normality: $p\text{-hat} = 19/172 \approx 0.11$, $q\text{-hat} = 0.89$. $np\text{-hat} = 172 \times 0.11 \approx 19 \geq 10$ and $nq\text{-hat} = 172 \times 0.89 \approx 153 \geq 10$ — condition satisfied.

4. Answer: $z^* = 1.960$

- $\alpha = 1 - 0.95 = 0.05$
- $\alpha/2 = 0.025$
- Using the standard normal table or calculator ($\text{invNorm}(0.025)$), $z^* = 1.960$.

5. Answer: (0.302, 0.378)

- $p\text{-hat} = 136/400 = 0.34$, $q\text{-hat} = 0.66$, $n = 400$.
- For 90% confidence, $z^* = 1.645$.
- Standard error = $\sqrt{0.34 \times 0.66 / 400} = \sqrt{0.0005610} \approx 0.02369$.
- Margin of error = $1.645 \times 0.02369 \approx 0.0390$.
- Confidence interval: $0.34 \pm 0.039 \rightarrow (0.301, 0.379) \approx (0.302, 0.378)$.

6. Answer: We are 95% confident that the true proportion of adults who support the new policy is between 0.52 and 0.68.

- A confidence interval gives a range of plausible values for the true population proportion.
- State the confidence level (95%), the parameter (proportion of adults supporting the policy), and the interval (0.52 to 0.68).
- Correct interpretation: We are 95% confident that the true proportion of adults who support the new policy is between 0.52 and 0.68.



**7. Answer: $n = 1068$**

- With no prior estimate, use $p\text{-hat} = 0.50$ and $q\text{-hat} = 0.50$ (most conservative).
- z^* for 95% confidence = 1.960, $E = 0.03$.
- $n = (1.960 / 0.03)^2 \times (0.50)(0.50) = (65.333)^2 \times 0.25 = 4268.4 \times 0.25 = 1067.1$.
- Round up: $n = 1068$.

8. Answer: $n = 318$ (round up from 317.1)

- Start with $E = z^* \times \sqrt{p\text{-hat} \times q\text{-hat} / n}$.
- Multiply both sides by $\sqrt{n}$: $E \times \sqrt{n} = z^* \times \sqrt{p\text{-hat} \times q\text{-hat}}$.
- Divide both sides by E : $\sqrt{n} = z^* \times \sqrt{p\text{-hat} \times q\text{-hat}} / E$.
- Square both sides: $n = (z^* \times \sqrt{p\text{-hat} \times q\text{-hat}} / E)^2$.
- Substitute: $z^* = 1.645$, $p\text{-hat} = 0.75$, $q\text{-hat} = 0.25$, $E = 0.04$.
- $\sqrt{n} = 1.645 \times \sqrt{0.1875} / 0.04 = 1.645 \times 0.4330 / 0.04 \approx 17.81$.
- $n = 17.81^2 \approx 317.1 \rightarrow$ round up to $n = 318$.

9. Answer: (0.637, 0.763); conditions are met; we are 99% confident the true proportion is between 0.637 and 0.763.

- $p\text{-hat} = 175/250 = 0.70$, $q\text{-hat} = 0.30$, $n = 250$.
- Conditions: SRS assumed; $N > 10(250) = 2500$ (reasonable for nurses); $np\text{-hat} = 175 \geq 10$ and $nq\text{-hat} = 75 \geq 10$. All satisfied.
- For 99% confidence: $\alpha/2 = 0.005$, $z^* = 2.576$.
- Standard error = $\sqrt{0.70 \times 0.30 / 250} = \sqrt{0.00084} \approx 0.02898$.
- Margin of error = $2.576 \times 0.02898 \approx 0.0747$.
- Confidence interval: $0.70 \pm 0.0747 \rightarrow (0.625, 0.775) \approx (0.637, 0.763)$.
- Interpretation: We are 99% confident that the true proportion of nurses experiencing burnout is between 0.637 and 0.763.

10. Answer: Study B has the narrower interval ($E \approx 0.027$) compared to Study A ($E \approx 0.081$) because a larger sample size reduces the standard error.

- Reasoning: Both studies have the same $p\text{-hat}$ and z^* , so the only difference is n . A larger n makes the standard error smaller, narrowing the margin of error.
- For both studies: $p\text{-hat} = 0.78$, $q\text{-hat} = 0.22$, $z^* = 1.960$.
- Study A ($n = 100$): $E = 1.960 \times \sqrt{0.78 \times 0.22 / 100} = 1.960 \times \sqrt{0.001716} = 1.960 \times 0.04142 \approx 0.0812$.
- Study B ($n = 900$): $E = 1.960 \times \sqrt{0.78 \times 0.22 / 900} = 1.960 \times \sqrt{0.0001907} = 1.960 \times 0.01381 \approx 0.0271$.
- Study B's margin of error (≈ 0.027) is much smaller than Study A's (≈ 0.081), confirming that a larger sample size produces a narrower confidence interval.

